

Security Tools Evaluation Report

Course: APDS7311wPOE

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1. Introduction

This report evaluates MobSF and ScoutSuite to determine their effectiveness in identifying security vulnerabilities in mobile applications and cloud environments.

2. MobSF (Mobile Security Framework)

MobSF is used to analyse mobile applications through static and dynamic analysis.

Key Findings:

- 1 Insecure network configuration allowing cleartext traffic
- 2 Weak cryptographic algorithms (MD5, SHA-1)
- 3 Hardcoded sensitive data
- 4 Insecure data storage
- 5 Insecure random number generator
- 6 Support for outdated Android versions

The application scored approximately 50/100, indicating moderate risk.

Security Assessment:

The APK is not secure for production use due to multiple vulnerabilities and risks such as data leakage and interception.

Advantages of MobSF:

- 1 Easy to use
- 2 Detailed reports
- 3 Industry standard alignment

- 4 Early vulnerability detection

Limitations of MobSF:

- 1 Static analysis focused
- 2 Possible false positives
- 3 Limited runtime simulation

3. ScoutSuite

ScoutSuite audits cloud environments such as AWS.

Typical Findings:

- 1 Misconfigured IAM roles
- 2 Publicly accessible storage
- 3 Weak security groups
- 4 Missing encryption
- 5 Logging issues

4. Recommendation

Both tools should be used together as part of a DevSecOps pipeline.

5. Conclusion

MobSF identified significant vulnerabilities, showing the APK is not safe for production. ScoutSuite complements by securing cloud infrastructure.